

```
    document.write(i + " Hello, world!<br />");
    i = i + 1;
}

document.write("Code in a function:<br />");
function hello(i) {
    if (i === undefined) {
        i = 0; // i defaults to 0 if unspecified.
    }

    document.write(i + " Hello, world!<br />");
    i = i + 1;
    if (i < 5) {
        hello(i); // RECURSIVE CASE
    }
    else {
        return; // BASE CASE
    }
}
hello();
</script>
```

The output of these programs looks like this:

```
Code in a loop:
0 Hello, world!
1 Hello, world!
2 Hello, world!
3 Hello, world!
4 Hello, world!
Code in a function:
0 Hello, world!
1 Hello, world!
2 Hello, world!
3 Hello, world!
4 Hello, world!
```

The `while` loop has a condition, `i < 5`, that determines whether the program keeps looping. Similarly, the recursive function uses this condition for its recursive case, which causes the function to call itself and execute the `hello, world!` to display its code again.